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ORIGINAL ARTICLE

On a set of J_2 equinoctial orbital elements and their use for uncertainty propagation

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Abstract
This paper defines a set of six non-singular orbital elements designed specifically for the characterization of uncertainty in the state of a resident space object in circular or elliptic orbit and demonstrates their use for uncertainty propagation in the context of the perturbed two-body problem of orbital mechanics. As evidenced by the time evolution of the Cramér–von Mises test statistic, representation of the orbital state probability density function in J_2EqOE yields less nonlinear uncertainty propagation and provides covariance and uncertainty realism for much longer periods of time than what is possible using Cartesian coordinates or even equinoctial orbital elements, without an appreciable increase in computational cost.

Keywords Satellites · Coordinate systems · Uncertainty propagation

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1 Introduction

Space situational awareness (SSA) is the comprehensive knowledge of the near-Earth space environment accomplished through the tracking and identification of orbiting space objects needed to protect man-made satellites and maintain awareness of potential collision hazards and on-orbit threats. Uncertainty realism is the proper characterization of the probability density function (PDF) of the state of a system.[1] Uncertainty (and covariance) realism in

[1] Covariance realism is the proper characterization of the covariance in the state of a system under Gaussian assumptions. It implies that the estimate of the mean is the true mean (i.e., the estimate is unbiased) and that the covariance possesses the right size, shape, and orientation (i.e., consistency). Uncertainty realism relaxes the Gaussian assumptions. Covariance realism is a necessary but not sufficient condition for achieving uncertainty realism. An extensive study on covariance and uncertainty realism in the SSA domain is provided in the report of Poore et al. (2016).

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the orbital state of a space object is a common requirement for many SSA functions, especially ones involving uncertainty propagation such as space surveillance tracking, conjunction assessment, sensor tasking, and maneuver detection (Aristoff et al. 2015b). Achieving uncertainty realism in the current or future state of a space object is challenging due to long time gaps between measurement updates, nonlinear and non-conservative dynamics, and long propagation times. One proven approach to improving covariance and uncertainty realism during uncertainty propagation is through the use of orbital-element-based coordinate systems that absorb the dominant nonlinearity in the equations of motion: the $1/r^2$ term in the geopotential. Of particular interest are non-singular orbital elements that prevent the singularities that exist for the standard orbital elements at zero eccentricity and zero inclination, such as the equinoctial orbital elements (Broucke and Cefola 1972). One might ask if one can do better; that is, can one absorb additional nonlinearities in the equations of motion to further improve uncertainty realism, while at the same time respecting constraints pertaining to implementation and runtime costs? The answer is a resounding yes.

This paper defines a set of orbital elements, coined the J_2 *equinoctial orbital elements* (J_2EqOE), designed specifically for the characterization of uncertainty in the state of a space object in circular or elliptic orbit, and demonstrates their use for uncertainty propagation in the context of the perturbed two-body problem of orbital mechanics. What distinguishes the J_2EqOE from existing sets of orbital elements is the combination of the following. First, the J_2EqOE are non-singular for eccentricities less than one and inclinations less than 180°. Hence, complications do not arise when propagating the uncertainty of nearly circular orbits or nearly equatorial orbits.[2] Second, the J_2EqOE account for the dominant perturbation to Earth-orbiting satellites, the J_2 zonal perturbation that arises from the oblateness of the Earth (Montenbruck and Gill 2005; Vallado 2013). Thus, when used for uncertainty propagation, the J_2EqOE enable one to maintain covariance and uncertainty realism for much longer periods of time (Horwood et al. 2014a, b). Third, the J_2EqOE are straightforward to implement. Fourth, the J_2EqOE are a set of only six non-singular orbital elements. Hence, unlike non-singular sets consisting of seven or more orbital elements, they do not require additional orbits to be propagated (e.g., when using sigma-point- or particle-based methods for uncertainty propagation) nor higher-dimensional state transition matrices to be numerically integrated (e.g., when using Kalman-based methods for uncertainty propagation). Thus, use of the J_2EqOE does not appreciably increase the computational cost of uncertainty propagation.

We note that the J_2EqOE can be described as a particular set of "mean" orbital elements. However, use of the J_2EqOE is not restricted to a particular force model nor orbital propagator. Thus, we prefer to view the J_2EqOE as simply a coordinate system in which the dynamics are less nonlinear, much like the osculating equinoctial elements (EqOE), but with the added benefit of having absorbed the J_2 perturbation. We also note that Gim and Alfriend (2005) developed a similar first-order J_2 analytic orbit theory using equinoctial elements and applied it to the problem of formation flying. We stress that use of the J_2EqOE in a Kalman filter does not require one to use an analytical propagation theory (e.g., general perturbations) in the prediction step. Indeed, the uncertainty propagation examples presented in this paper use numerical integration based on the authors' Gauss–Legendre implicit Runge–Kutta orbital propagator (Aristoff et al. 2014a, 2015a) replete with high fidelity force models.

[2] More than 25% of cataloged space objects have eccentricities less than 10^{-3} and nearly 2% have eccentricities less than 10^{-4} (Space-Track.Org 2020).

The primary application of this work is the uncertainty propagation (filter prediction) problem. We acknowledge that the filter correction step is equally important in many SSA use cases. However, there are applications where preserving the Gaussian assumption in the prediction step of a Kalman filter is advantageous (i.e., by representing the Gaussian PDF in an orbital element coordinate system), most notably, in the problem of estimating the probability of collision (PC) between two objects. As argued in Aristoff et al. (2014b), it is more efficient to propagate the states and covariances representing the two objects in (osculating or J_2) equinoctial orbital elements to the time of closest approach. Then, if needed, the propagated Gaussian PDFs can be converted to Gaussian mixtures in Cartesian coordinates to facilitate the PC calculation (which is inherently defined in Cartesian coordinates). This procedure is computationally less costly than propagating an entire Gaussian mixture in Cartesian coordinates from epoch to the time of closest approach (since propagating a Gaussian PDF under the nonlinear dynamics is generally more expensive than converting a Gaussian PDF from element space to Cartesian space). This paradigm is followed in the KRATOS PC algorithm (Horwood et al. 2016). An analogous procedure is applicable to nonlinear filtering problems in SSA. In the implementation of the filter correction step, one can convert the propagated orbital element Gaussian PDF to a Gaussian mixture in Cartesian coordinates, if the nonlinearities in the state-to-measurement function are sufficiently high to warrant such a conversion, otherwise one can apply the conventional (extended or unscented) Kalman filter update.

The layout of this paper is the following. In Sect. 2, we define the coordinate transform between EqOE and J_2EqOE. An algorithm for inverting the transform is also presented. In Sect. 3, we illustrate the time evolution of the J_2EqOE, and demonstrate the use of the J_2EqOE for uncertainty propagation using a range of initial conditions. In Sect. 4, we discuss potential extensions and alternatives to the J_2EqOE for the purposes of improved uncertainty propagation. Section 5 concludes our study. In the appendix, we provide a numerical example of the J_2EqOE to EqOE conversion, inclusive of all intermediate steps, to serve as a test case and debugging tool for readers who may want to implement the transformations.

2 Methods and theory

This section defines the coordinate transform to and from the J_2EqOE and the (osculating) EqOE that is valid (i.e., non-singular) for eccentricities less than one and for inclinations less than 180°, including orbits with zero eccentricity or zero inclination, following the work of Brouwer (1959) and Lyddane (1963). At a high level, the transformation to and from J_2EqOE and EqOE takes the form below.

$$\begin{array}{c} \text{J}_2\text{EqOE} \\ (n'', h'', k'', p'', q'', \ell'') \end{array} \overset{\text{Sect. 2.1}}{\overline{\qquad\qquad}} \begin{array}{c} \text{J}_2\text{IOE} \\ (I_1'', \ldots, I_6'') \end{array} \overset{\text{Sect. 2.2}}{\overline{\qquad\qquad}} \begin{array}{c} \text{IOE} \\ (I_1, \ldots, I_6) \end{array} \overset{\text{Sect. 2.1}}{\overline{\qquad\qquad}} \begin{array}{c} \text{EqOE} \\ (a, h, k, p, q, \ell) \end{array}$$

Here, the IOE are combinations of the Keplerian orbital elements defined by Lyddane (1963) that we coin the *intermediate orbital elements*. We use the IOE as an intermediate step in the transformation to avoid singularities at zero eccentricity and/or zero inclination that are present in Lyddane's formulation,[3] as well as in Brouwer's formulation.[4] Labels above the lines point to sections in which the individual transformations are defined.

[3] Lyddane's algorithm contains an intrinsic singularity because the final output is the Keplerian elements.

[4] Brouwer's algorithm contains explicit singularities for zero eccentricity and zero inclination.

2.1 EqOE to/from IOE

The equinoctial orbital elements (Broucke and Cefola 1972) are a system of six non-singular elements which are suited for all types of (elliptic) orbits, including those with low eccentricities and inclinations. The equinoctial elements (a, h, k, p, q, ℓ) are defined in terms of the Keplerian orbital elements $(a, e, i, \Omega, \omega, M)$ by:

$$\begin{aligned} &a = a, \quad h = e\sin(\omega + \Omega), \quad k = e\cos(\omega + \Omega), \\ &p = \tan\tfrac{i}{2}\sin\Omega, \quad q = \tan\tfrac{i}{2}\cos\Omega, \quad \ell = M + \omega + \Omega, \end{aligned} \tag{1}$$

where a is the semi-major axis, e is the eccentricity, i is the inclination, Ω is the right ascension of the ascending node, ω is the argument of periapsis, and M is the mean anomaly. The intermediate orbital elements $(I_1, I_2, I_3, I_4, I_5, I_6)$ are defined in terms of the Keplerian orbital elements (KepOE) by:

$$\begin{aligned} &I_1 = a, \quad I_2 = e\sin(M), \quad I_3 = e\cos(M), \\ &I_4 = \sin\tfrac{i}{2}\sin\Omega, \quad I_5 = \sin\tfrac{i}{2}\cos\Omega, \quad I_6 = M + \omega + \Omega. \end{aligned} \tag{2}$$

Conversion from IOE to KepOE is given by:

$$\begin{aligned} &a = I_1, \quad e = \sqrt{I_2^2 + I_3^2}, \quad i = 2\sin^{-1}\sqrt{I_4^2 + I_5^2}, \\ &\Omega = \tan^{-1}(I_4, I_5), \quad \omega = I_6 - M - \Omega, \quad M = \tan^{-1}(I_2, I_3). \end{aligned} \tag{3}$$

Note that the two-argument inverse tangent is used when computing Ω and M above, and that $\tan^{-1}(0, 0) \equiv 0$. Conversion from EqOE to IOE is given by:

$$\begin{aligned} &I_1 = a, \quad I_2 = k\sin(\ell) - h\cos(\ell), \quad I_3 = h\sin(\ell) + k\cos(\ell), \\ &I_4 = \tfrac{p}{\sqrt{1+p^2+q^2}}, \quad I_5 = \tfrac{q}{\sqrt{1+p^2+q^2}}, \quad I_6 = \ell. \end{aligned} \tag{4}$$

Conversion from IOE to EqOE is given by:

$$\begin{aligned} &a = I_1, \quad h = I_3\sin(I_6) - I_2\cos(I_6), \quad k = I_2\sin(I_6) + I_3\cos(I_6), \\ &p = \tfrac{I_4}{\sqrt{1-I_4^2-I_5^2}}, \quad q = \tfrac{I_5}{\sqrt{1-I_4^2-I_5^2}}, \quad \ell = I_6. \end{aligned} \tag{5}$$

The alternative EqOE coordinate system (AEqOE) is one in which orbit propagation becomes linear for the case of unperturbed Keplerian dynamics. The difference between EqOE and AEqOE is that the first element is not the semi-major axis a, but rather the mean motion $n = \sqrt{\mu/a^3}$, where $\mu = GM$ is the gravitational parameter. For the J_2EqOE, we use this alternative definition of the EqOE, and apply this when converting J_2EqOE to/from J_2IOE (with all unprimed variables replaced by their double-primed counterparts).

2.2 J_2IOE to/from IOE

In this section, we define a J_2IOE to IOE transformation by considering only short-period terms from the Hamiltonian potential equation, expanded in a perturbation series up to order J_2, following Brouwer (1959) and Lyddane (1963). We seek a definition that is valid for all eccentricities less than one, and for all inclinations less than 180°. Therefore, we do not include the long-period terms as they possess a singularity at the critical inclinations of approximately 63.4° and 116.6°. Fortunately, long-period terms (which are functions of the "slowly" varying argument of perigee through trigonometric relations) can be safely

neglected because propagation times are typically less than 1 week in the aforementioned SSA functions.

In order to be consistent with the notation used by Brouwer (1959), Lyddane (1963), and Cain (1962), we denote the Keplerian elements as (a, e, I, h, g, l) rather than $(a, e, i, \Omega, \omega, M)$ in this section. Therefore, h denotes the right ascension of the ascending node, not the second equinoctial orbital element as in Sect. 2.1.

2.2.1 J_2IOE to IOE

The transformation from J_2IOE to IOE consists of nine steps. Note that double-primed quantities refer to J_2 elements, and we drop the subscript from k_2, γ_2 and γ_2' in Step 2 since we do not consider higher-order harmonics in the geopotential.

Step 1 Compute the Keplerian orbital elements $(a'', e'', I'', h'', g'', l'')$ from the J_2IOE via (3). Note that although the Keplerian elements are used in the transformation, the osculating elements l, g, and h are never individually calculated, only their sum, later in Step 6. Since this sum is the mean longitude variable in the EqOE definition, it and its derivatives are non-singular, and the intrinsic singularity present in Lyddane's formulation is not encountered.

Step 2 Compute the auxiliary quantities $k = \frac{1}{2} J_2 R^2$, $\eta = \sqrt{1 - e''^2}$, $\gamma = k/a''^2$, $\gamma' = \gamma/\eta^4$, and $\theta = \cos I''$, where R is the radius of the Earth, $\mu = GM$ is the gravitational parameter, and J_2 is the un-normalized J_2 parameter (i.e., the $C_{2,0}$ geocoefficient).

Step 3 Solve Kepler's equation $E'' - e'' \sin E'' = l''$ to find the eccentric anomaly E''. It is recommended that this transcendental equation be solved using the Newton–Raphson method, followed by the bisection method to ensure convergence of the iterations, using a sufficiently high tolerance.

Step 4 Compute the radius (distance from object to focus) r'' and the true anomaly f'' via

$$\begin{aligned} r'' &= a''\left(1 - e'' \cos E''\right), \\ f'' &= \tan^{-1}\left(\eta \sin E'', \cos E'' - e''\right), \end{aligned} \tag{6}$$

and perform a branch-cut correction[5] on f'' and l'' as l'' will be subtracted from f'' in Steps 5 and 6.

Step 5 Compute a and δh via

$$\begin{aligned} a &= a''\left\{1 + \gamma\left[\left(-1 + 3\theta^2\right)\left(\frac{a''^3}{r''^3} - \eta^{-3}\right) + 3\left(1 - \theta^2\right)\frac{a''^3}{r''^3}\cos(2g'' + 2f'')\right]\right\}, \\ \delta h &= -\frac{1}{2}\gamma'\theta\Big[6\left(f'' - l'' + e'' \sin f''\right) - 3\sin\left(2g'' + 2f''\right) \\ &\quad - 3e'' \sin\left(2g'' + f''\right) - e'' \sin\left(2g'' + 3f''\right)\Big]. \end{aligned} \tag{7}$$

Step 6 Compute $l + g + h$ via

$$\begin{aligned} l + g + h = \frac{1}{4}\gamma'\Big\{&6\left(-1 + 5\theta^2\right)\left(f'' - l'' + e'' \sin f''\right) + \left(3 - 5\theta^2\right) \\ &\times\left[3\sin\left(2g'' + 2f''\right) + 3e'' \sin\left(2g'' + f''\right) + e'' \sin\left(2g'' + 3f''\right)\right]\Big\} \\ &+ \delta h + l'' + g'' + h''. \end{aligned} \tag{8}$$

[5] Suppose θ_1 and θ_2 are two angles equivalent up to integer 2π shifts. Let $b_{cut} = \theta_1 - \pi$, and redefine θ_2 according to $\theta_2 = \theta_2 + 2\pi\lceil(b_{cut} - \theta_2)/(2\pi)\rceil$. Then, θ_1 and θ_2 can be safely added or subtracted.

Step 7 Compute δe and $e''\delta l$ via

$$\begin{aligned}\delta e &= \frac{1}{2}\eta^2\Big\{\gamma\big[\left(-1+3\theta^2\right)v_2 + 3\left(1-\theta^2\right)v_3\cos\left(2g''+2f''\right)\big] \\ &\quad - \gamma'\left(1-\theta^2\right)\big[3\cos\left(2g''+f''\right)+\cos\left(2g''+3f''\right)\big]\Big\}, \\ e''\delta l &= -\frac{1}{4}\eta^3\gamma'\Big\{2\left(-1+3\theta^2\right)(v_4+1)\sin f'' + 3\left(1-\theta^2\right) \\ &\quad \times\left[(-v_4+1)\sin\left(2g''+f''\right)+\left(v_4+\frac{1}{3}\right)\sin\left(2g''+3f''\right)\right]\Big\},\end{aligned} \tag{9}$$

where we have used the identities suggested by Lyddane (1963) and where

$$\begin{aligned}&v_1 = 3\cos f'' + 3e''\cos^2 f'' + e''^2\cos^3 f'', \quad v_2 = \eta^{-6}\left(e''\eta + \frac{e''}{1+\eta} + v_1\right), \\ &v_3 = \eta^{-6}\left(e''+v_1\right), \quad v_4 = \left(\frac{a''\eta}{r''}\right)^2 + \frac{a''}{r''}.\end{aligned} \tag{10}$$

Step 8 Compute δI and $\sin\left(\frac{1}{2}I''\right)\delta h$ via

$$\begin{aligned}\delta I &= \frac{1}{2}\gamma'\theta\sqrt{1-\theta^2}\big[3\cos\left(2g''+2f''\right)+3e''\cos\left(2g''+f''\right)+e''\cos\left(2g''+3f''\right)\big], \\ \sin\left(\tfrac{1}{2}I''\right)\delta h &= \frac{\sin(I'')\delta h}{2\cos(\frac{1}{2}I'')}.\end{aligned} \tag{11}$$

Step 9 Compute the IOE via

$$\begin{aligned}I_1 &= a, \\ I_2 &= (e''+\delta e)\sin l'' + e''\delta l\cos l'', \\ I_3 &= (e''+\delta e)\cos l'' - e''\delta l\sin l'', \\ I_4 &= \left[\sin\left(\tfrac{1}{2}I''\right)+\cos\left(\tfrac{1}{2}I''\right)\tfrac{1}{2}\delta I\right]\sin h'' + \sin\left(\tfrac{1}{2}I''\right)\delta h\,\cos h'', \\ I_5 &= \left[\sin\left(\tfrac{1}{2}I''\right)+\cos\left(\tfrac{1}{2}I''\right)\tfrac{1}{2}\delta I\right]\cos h'' - \sin\left(\tfrac{1}{2}I''\right)\delta h\,\sin h'', \\ I_6 &= l+g+h.\end{aligned} \tag{12}$$

Finally, we note that by computing the Jacobian directly, one can verify that the J_2IOE to IOE transformation is non-singular. However, we omit details for the sake of brevity.

2.2.2 IOE to J_2IOE

The transformation from IOE to J_2IOE requires inverting the J_2IOE to IOE transform defined in Sect. 2.2.1. Specifically, it requires solving six coupled nonlinear equations which map the J_2IOE to the IOE, similar to how one inverts the transform introduced by Brouwer (Cain 1962). Note that we seek a numerical solution that does not require computation of the Jacobian of the J_2IOE to IOE transform. A careful implementation of fixed-point iteration satisfies this criterion.

Four considerations are necessary for a robust solution of the nonlinear system of equations using fixed-point iteration. First, we iterate over the scaled J_2IOE to improve the conditioning of the fixed-point iterations. Second, we treat the angular variables via a branch-cut correction.

Table 1 Initial Keplerian orbital elements for the examples in Sect. 3

Scenario	Orbit type	a (km)	e	i (°)	Ω (°)	ω (°)	M (°)
1	LEO	7136.6	0.00949	72.9	116.0	57.7	105.5
2	GEO	42164.1	0.0	0.0	0.0	0.0	250.0
3	HEO	26628.1	0.742	63.4	120.0	0.0	144.0

Third, we enforce constraints on the J_2IOE. Fourth, we use an initial guess that enables convergence in typically under five iterations. A description of the algorithm is given below. Once implemented, round-trip tests of the coordinate system are recommended.

Step 1 Input the IOE and let the initial guess for the J_2IOE be these elements. If any of the IOE are less than 0.01, increase them to 0.01 to improve convergence.

Step 2 While the two-norm of the residuals is greater than the iteration tolerance and the number of iterations is less than the maximum allowable number of iterations, do Steps 2a–2d.

Step 2a Compute the IOE from the J_2IOE guess using the transformation defined in Sect. 2.2.1 and scale the result by $(R, 1, 1, 1, 1, 1)$, where R is the radius of the Earth.

Step 2b Perform a branch-cut correction on the sixth IOE with respect to the sixth J_2IOE, and compute the residuals between the (scaled) input J_2IOE and those found in Step 2a.

Step 2c Compute the two-norm of the residuals found in Step 2b. If the norm is lower than all previously computed residual norms (or it is the first iteration), record the current (scaled) IOE (i.e., the best solution found thus far).

Step 2d Compute the next guess for the (scaled) IOE by subtracting the residuals from the current guess. Enforce the constraint that I_2, I_3, I_4, and I_5 must be between -1 and 1.

Step 3 Return the J_2IOE corresponding to the lowest residual norm.

3 Results and discussion

This section provides a number of applications of the coordinate transform defined in Sect. 2. In particular, we consider the time evolution of the J_2EqOE in Sect. 3.1 and their use for uncertainty propagation in Sect. 3.2. The initial (Keplerian) orbital elements for the examples are given in Table 1. Numerical integration of the equations of motion, i.e., to compute the ephemerides in Sect. 3.1 and to propagate particles and sigma-points in Sect. 3.2, is performed using the authors' variable-step Gauss–Legendre implicit Runge–Kutta orbital propagator (Aristoff et al. 2014a, 2015a).

The dynamical models used in this testing are comprised of the following forces. For the object in low Earth Orbit (LEO), a degree and order 32 Earth gravity model is used.[6] For the object in geostationary Earth orbit (GEO), a degree and order 8 Earth gravity model is used, together with lunar–solar perturbations[7] and a cannonball solar radiation pressure model with

[6] The geocoefficients are taken from the 2008 Earth gravity model (Pavlis et al. 2008).

[7] High-precision ephemerides (DE405) from the Jet Propulsion Laboratory's HORIZONS system are used for determining the position of the Sun and Moon.

a nominal ballistic coefficient of 0.1 m^2/kg. For the object in highly elliptic orbit (HEO), a degree and order 32 Earth gravity model is used, together with lunar–solar perturbations and a nominal ballistic coefficient of 0.1 m^2/kg. In the captions of the subsequent figures, we refer to these models as "full dynamics." An epoch of 2008-01-01T00:00:00Z is applied in each example.

3.1 Time evolution of the J_2EqOE

In this section, we validate the coordinate transformations by numerically propagating the orbits listed in Table 1 and plotting the AEqOE and J_2EqOE states at every 1/1000th of an orbit for three orbital periods. The purpose of this exercise is to verify that the time variation of the J_2EqOE is less nonlinear than that of the EqOE and that no biases are observed between the element sets for a given orbital element, where we have converted the semi-major axis to mean motion for the sake of comparison. Two different force models are used to represent truth: (i) a force model consisting of only J_2 gravity, and (ii) the "full dynamics" model that includes the additional perturbations described above. Recall that the J_2EqOE attempt to "absorb" the J_2 perturbation, and that higher-order gravitational perturbations as well as solar-lunar perturbations and other forces are omitted. The examples wherein we use a higher-fidelity "truth" to model the dynamics will allow us to quantify the effect of these omissions, and give us some indication of the potential benefit when using the J_2EqOE for uncertainty propagation.

Figure 1 illustrates the coordinate transform for an object in LEO. As expected the J_2EqOE exhibit less time variation than the AEqOE, and no bias is observed. Similar results are obtained for both force models, indicating that the J_2 term is by far the most dominant contributor to the perturbations in this example (Scenario 1).

Figure 2 illustrates the coordinate transform for an object in GEO. Again, the J_2EqOE tend to dampen the oscillations observed in the orbital elements, albeit less so when the higher-fidelity force model is used. Here, solar radiation pressure, and to a lesser extent the higher-order gravitational perturbations, causes the noticeable difference between the left and right columns; however, the magnitude of the oscillations is much less than what is observed in Fig. 1 for the low Earth orbit. Note that the oscillation in the J_2EqOE observed in orbital elements p and q for J_2 dynamics is negligible (below machine precision) and arises from the fact that the inclination is zero in this example (Scenario 2), which also serves to validate the coordinate transform for satellites with zero eccentricity and/or zero inclination.

Figure 3 illustrates the coordinate transform for an object in HEO. Near perigee, the AEqOE exhibit large-amplitude oscillations, especially in the mean motion, that are largely mitigated by use of the J_2EqOE. This is true for both force models under consideration. Similar results are obtained for both force models, indicating that the J_2 term is also the most dominant contributor to the perturbations.

In all three examples provided, use of the J_2EqOE resulted in a time variation of the orbital elements that was less nonlinear (i.e., more dampened oscillations). This was especially apparent in the LEO and HEO test cases, and suggests that covariance and uncertainty realism may be easier to maintain when the orbital state PDF is represented in J_2EqOE as opposed to AEqOE or EqOE. Additional examples are provided in the next section to demonstrate this fact.

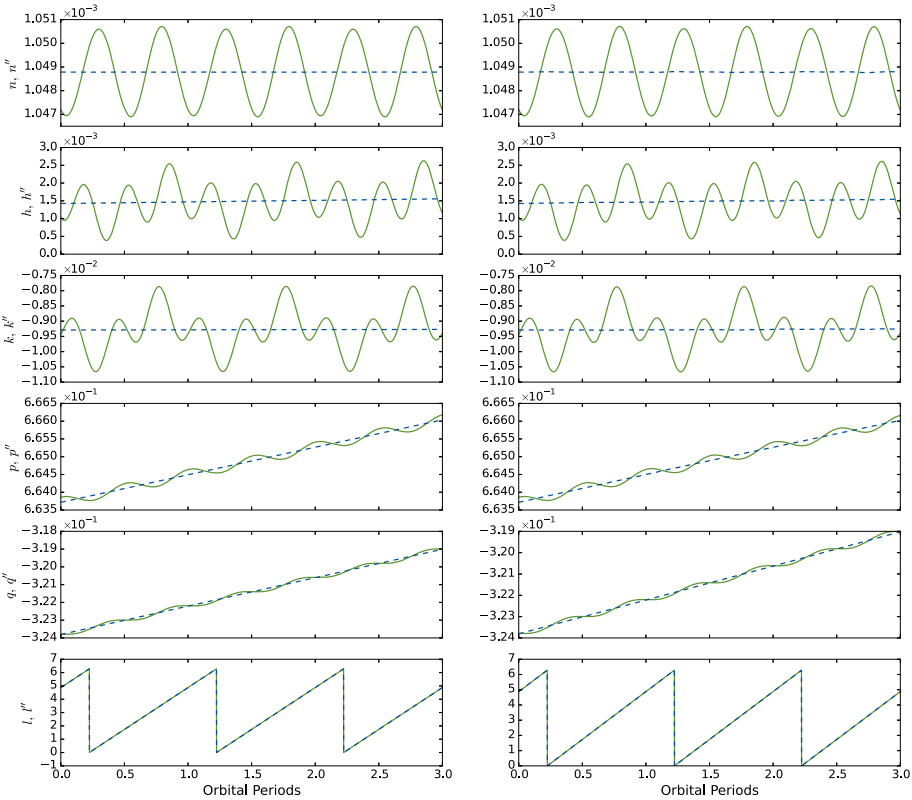


Fig. 1 Time evolution of the osculating AEqOE (solid green curves) and J_2EqOE (dashed blue curves) for an object in LEO (Scenario 1) that is numerically propagated using J_2 dynamics (left column) and full dynamics (right column)

3.2 Use of the J_2EqOE for uncertainty propagation

The purpose of this section is to demonstrate the use of the J_2EqOE for uncertainty propagation and to compare this to existing approaches in which the orbital state PDF is represented in EqOE or Cartesian coordinates. The actual testing is an off-line process conducted using multiple Monte Carlo trials. Specifically, we select a coordinate system in which to represent the orbital state and covariance at epoch, and we propagate the uncertainty to a future time using the prediction step of a nonlinear filter.[8] We then randomly sample points from the initial distribution and propagate each point to the same time. Next, we compute the Mahalanobis distance between the ith sample point, $\boldsymbol{x}^{(i)}$, and the filter-produced PDF.[9] Finally, we quantify the goodness-of-fit of the distribution of Mahalanobis distances to the expected

[8] Note that the orbital propagations themselves are performed in Cartesian space; however, the PDF need not be represented in Cartesian coordinates.

[9] The Mahalanobis distance is given by

$$\mathcal{M}^{(i)}(\boldsymbol{x}^{(i)}; \boldsymbol{\mu}, \mathbf{P}) = (\boldsymbol{x}^{(i)} - \boldsymbol{\mu})^T \mathbf{P}^{-1} (\boldsymbol{x}^{(i)} - \boldsymbol{\mu}), \tag{13}$$

where μ is the n-dimensional state estimate, and $\mathbf{P}$ the n by n covariance matrix. The expected value of $\mathcal{M}$ is n, and it follows that $\mathcal{M} \sim \chi^2(n)$, i.e., $\mathcal{M}$ is chi-squared distributed with n degrees of freedom.

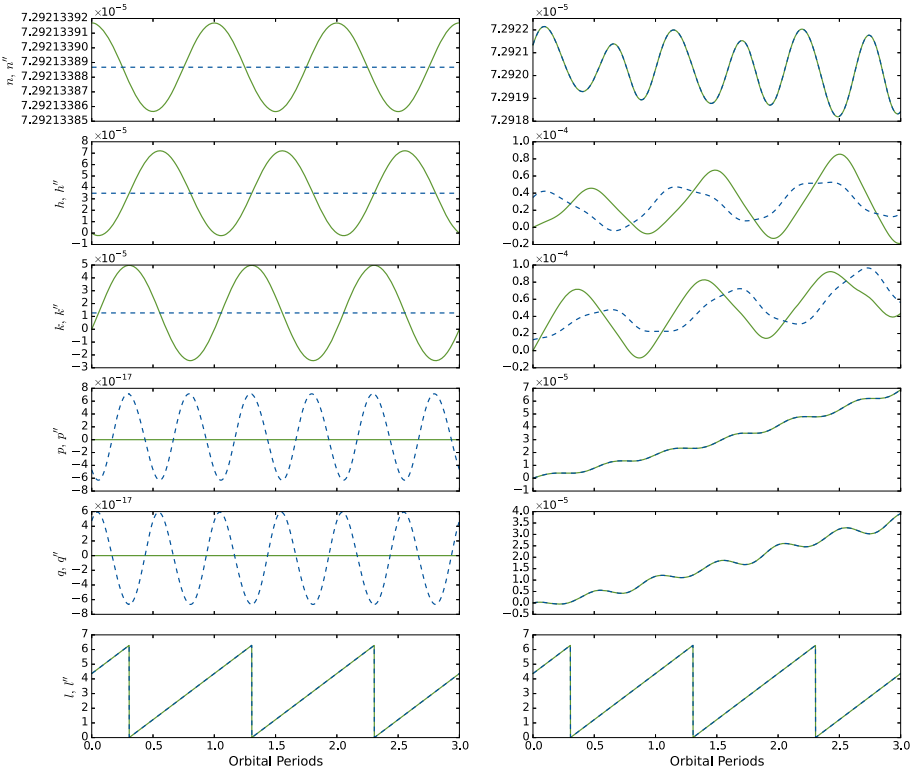


Fig. 2 Time evolution of the osculating AEqOE (solid green curves) and J_2EqOE (dashed blue curves) for an object in GEO (Scenario 2) that is numerically propagated using J_2 dynamics (left column) and full dynamics (right column)

chi-squared distribution using the Cramér–von Mises (CvM) test statistic.[10] By repeating this test for multiple time points over the propagation interval, we can identify when uncertainty (and covariance) realism breaks down during propagation given a particular set of initial conditions and filter configuration (and confidence level). Full details of this testing procedure are given in Horwood et al. (2014a). Here, we consider both the prediction step of the extended Kalman filter (Jazwinski 1970) and the unscented Kalman filter (Julier et al. 2000), two of the most commonly used methods for uncertainty propagation in space surveillance, also known as linearized propagation (LP) and the unscented transform (UT), respectively.

The initial Keplerian orbital elements for the examples considered in this testing are listed in Table 1. The covariances (in EqOE) are listed in the subsequent figure captions. To obtain the epoch Gaussian PDF where the mean and covariance are in a common coordinate system, the unscented transform was applied (Julier et al. 2000). The resulting Gaussian PDF

[10] The Cramér–von Mises test statistic is given by

$$Q = \frac{1}{12k} + \sum_{i=1}^{k} \left[\frac{2i-1}{2k} - F(y^{(i)}) \right]^2, \tag{14}$$

where the $y^{(i)}, i = 1, \ldots, k$, are the observed samples in increasing order, and F is the cumulative distribution function of the target distribution. Further details are given in Horwood et al. (2014b).

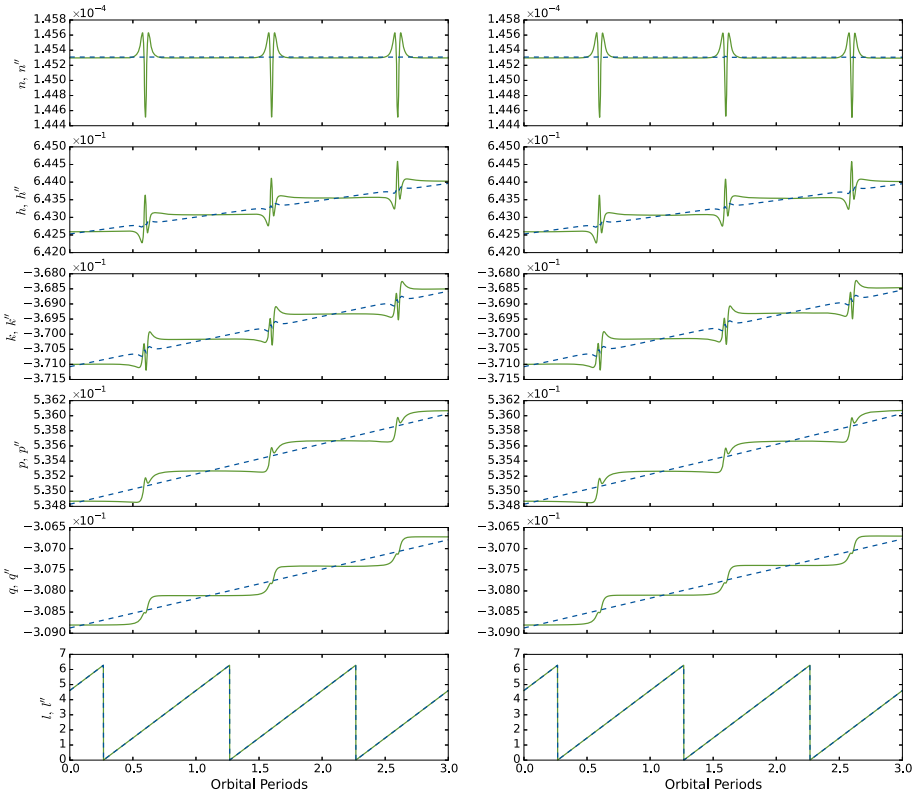


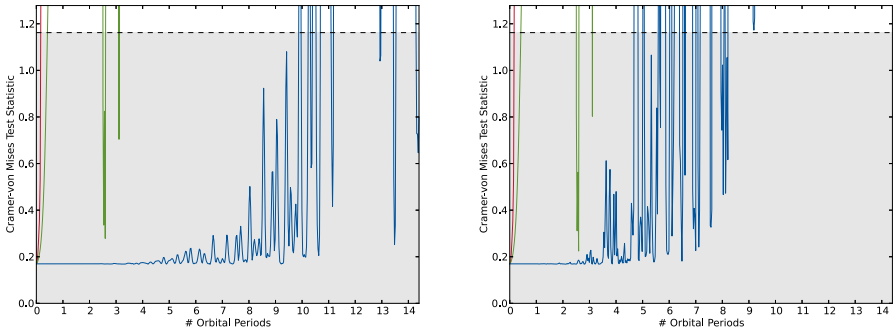
Fig. 3 Time evolution of the osculating AEqOE (solid green curves) and J_2EqOE (dashed blue curves) for an object in HEO (Scenario 3) that is numerically propagated using J_2 dynamics (left column) and full dynamics (right column)

was regarded as "truth" for the purpose of initiating the Monte Carlo samples. Nonetheless, the authors verified that covariance realism was preserved under these nonlinear coordinate transformations (details omitted).

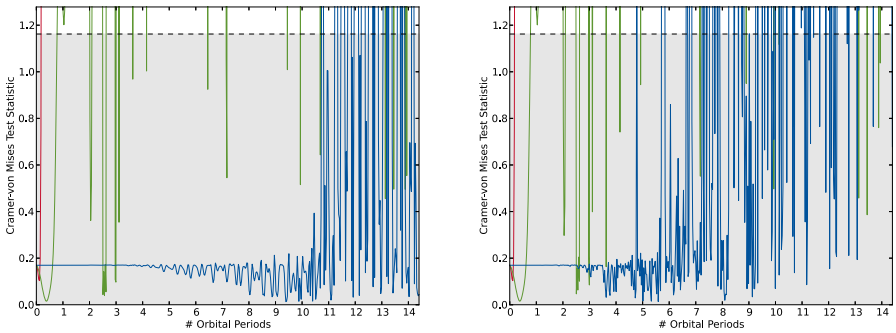
The uncertainty propagation test results, using 10,000 Monte Carlo trials to assess uncertainty realism (at 500 equally spaced time points), are presented in Figs. 4, 5, 6, and 7. Each figure contains four subfigures. Subfigures in a particular row pertain to a fixed uncertainty propagation method (LP or UT). Subfigures in a particular column pertain to a fixed force model. In each subfigure, we consider the influence of the coordinate system used to represent the PDF on the Cramér–von Mises test statistic. The shaded regions denote the interval above which uncertainty realism breaks down according to a 99.9% confidence level. The colors denote the different coordinate systems: red for Cartesian ECI coordinates,[11] green for EqOE, and blue for J_2EqOE. A number of observations can be made.

1. In all cases, using Cartesian coordinates to represent the PDF yields poor performance, regardless of whether the LP or UT is used, and regardless of which force model is

[11] Conceivably, the tests described in this section could be performed in other systems of Cartesian coordinates, such as the perifocal PQW coordinates or the RSW or NTW satellite coordinate systems (Vallado 2013). However, because any two Cartesian frames are related by an affine transformation (i.e., a rotation and translation), the same results would be obtained if one used a different Cartesian coordinate system.



(a) LP: CvM test statistic with J_2 dynamics **(b)** LP: CvM test statistic with full dynamics

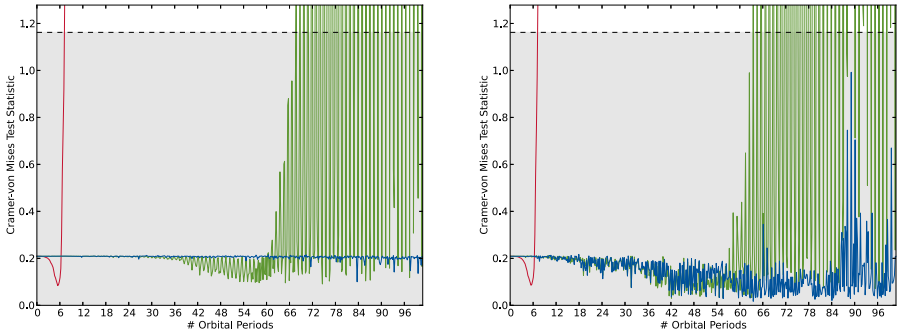


(c) UT: CvM test statistic with J_2 dynamics **(d)** UT: CvM test statistic with full dynamics

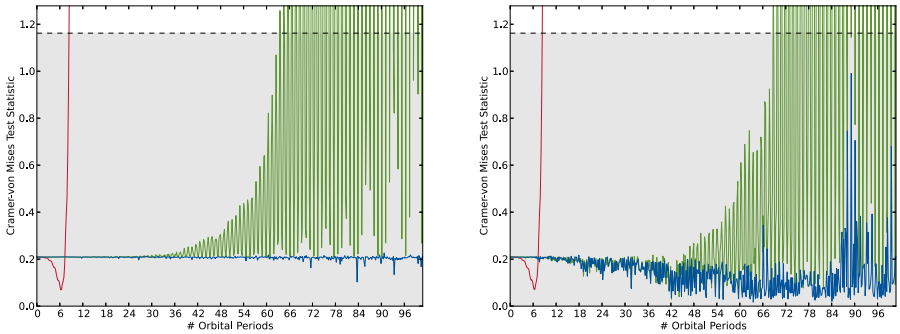
Fig. 4 Statistical measure of uncertainty realism applied to the LP and the UT for an object in LEO (Scenario 1) with initial EqOE covariance matrix $\mathbf{P} = \mathbf{A}\mathbf{A}^T$, where $\mathbf{A} = \text{diag}(\sigma_a = 20\,\text{km}, \sigma_h = 10^{-3}, \sigma_k = 10^{-3}, \sigma_p = 10^{-3}, \sigma_q = 10^{-3}, \sigma_\ell = 36'')$. The orbital state PDF is represented in Cartesian coordinates (red), EqOE (green), and J_2EqOE (blue), and is propagated for 1 day. For the PDF to be realistic (with 99.9% confidence), it is necessary that the value of the metric be within the shaded region

used. Uncertainty realism can be substantially improved by instead using EqOE. Further improvements can be made by using J_2EqOE to represent uncertainty. Specifically, as first reported in Horwood et al. (2014a) and corroborated by the results in this section, *uncertainty realism can be maintained up to 28 times longer by using the J_2EqOE in place of Cartesian coordinates, and up to 14 times longer by using the J_2EqOE in place of the* (Broucke and Cefola 1972) *osculating EqOE.* Importantly, doing so has negligible impact on the overall computational cost of uncertainty propagation.

2. The omission of higher-order perturbations (beyond J_2) has a negligible effect on the breakdown of uncertainty realism when the PDF is represented in Cartesian coordinates or in EqOE. Conversely, omission of higher-order perturbations extends the life of the UT and the LP with respect to uncertainty realism when the PDF is represented in J_2EqOE.
3. For a given coordinate system used to represent the orbital state PDF, uncertainty realism is improved (relative to the LP) by using the UT.
4. For the GEO test case, representation of uncertainty in EqOE and J_2EqOE give similar results, and the use of higher-order perturbations has a negligible effect on the evolution of the test statistic.



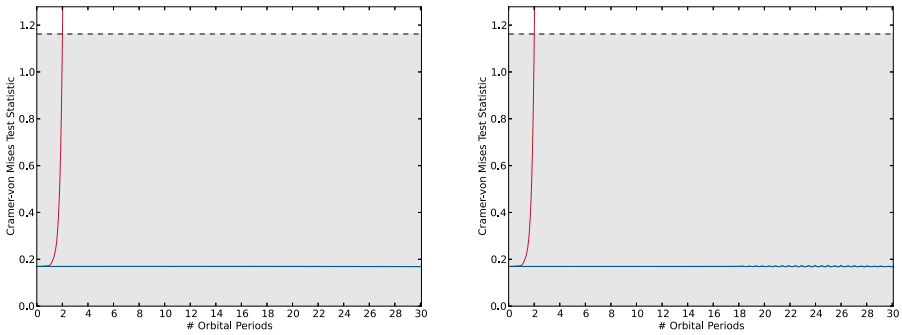
(a) LP: CvM test statistic with J_2 dynamics **(b)** LP: CvM test statistic with full dynamics



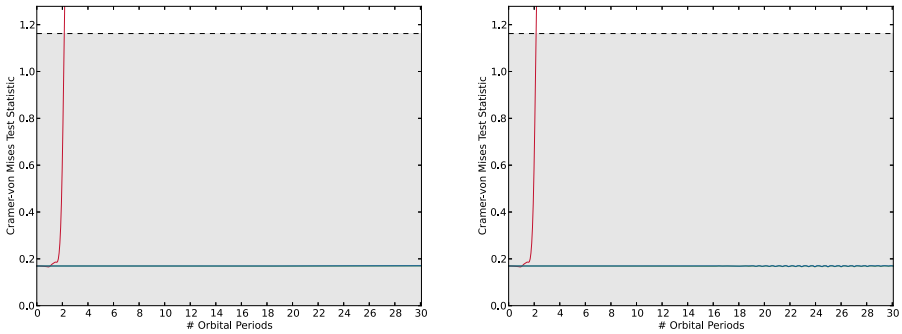
(c) UT: CvM test statistic with J_2 dynamics **(d)** UT: CvM test statistic with full dynamics

Fig. 5 Statistical measure of uncertainty realism applied to the LP and the UT for an object in LEO (Scenario 1) with initial EqOE covariance matrix $\mathbf{P} = \mathbf{A}\mathbf{A}^T$, where $\mathbf{A} = \text{diag}(\sigma_a = 50\,\text{m}, \sigma_h = 10^{-5}, \sigma_k = 10^{-5}, \sigma_p = 10^{-5}, \sigma_q = 10^{-5}, \sigma_\ell = 20'')$. The orbital state PDF is represented in Cartesian coordinates (red), EqOE (green), and J_2EqOE (blue), and is propagated for 7 days. For the PDF to be realistic (with 99.9% confidence), it is necessary that the value of the metric be within the shaded region

5. For the HEO test case, spikes in the CvM test statistic coincide with the space object being near perigee where the dynamics are most nonlinear. For the GEO test case, uncertainty realism is maintained throughout the 30-day propagation when the PDF is represented in EqOE or J_2EqOE.
6. The prediction step of each filter configuration eventually breaks down, as it should due to the nonlinearities in the dynamics. Measurement updates are necessary to avoid breakdown of uncertainty realism.
7. There are cases wherein the CvM test statistic is within the confidence region at a later time, but outside of the confidence region at an earlier time. Hence, if one were to check the performance of an uncertainty propagation method at only the later time, one could mistakenly believe that uncertainty realism had been maintained throughout the entire propagation. It is therefore crucial to quantify the *time evolution* of the CvM test statistic.
8. For a given uncertainty propagation method and scenario configuration at epoch, the values of the CvM test statistic are indistinguishable between the different coordinate systems. This indicates that nonlinearities in the coordinate transformations do not lead to loss of information at the *initial time* in these examples.



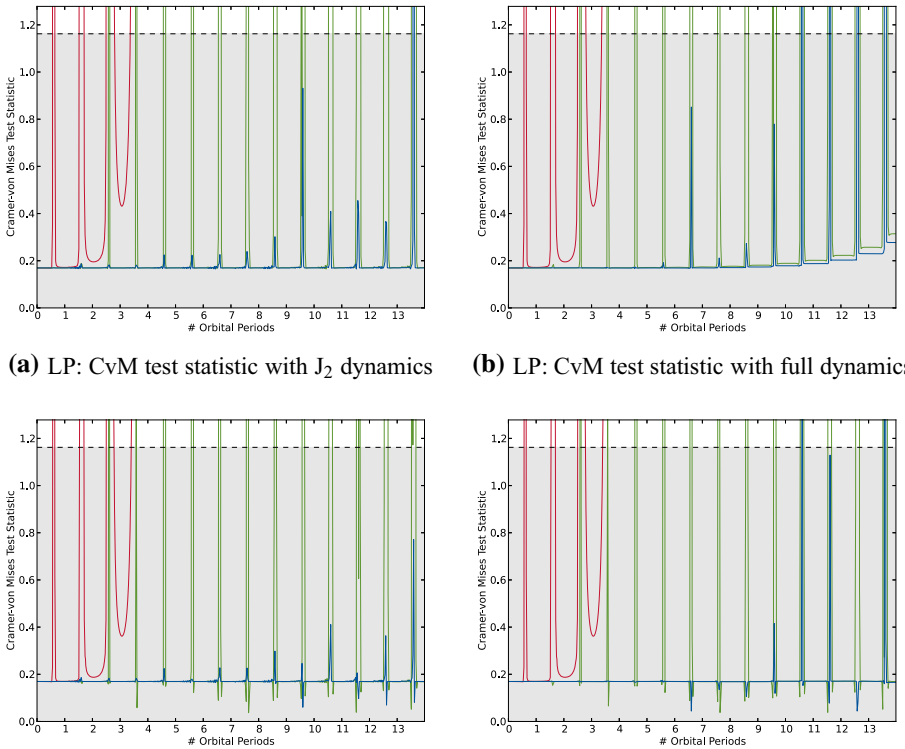
(a) LP: CvM test statistic with J_2 dynamics **(b)** LP: CvM test statistic with full dynamics



(c) UT: CvM test statistic with J_2 dynamics **(d)** UT: CvM test statistic with full dynamics

Fig. 6 Statistical measure of uncertainty realism applied to the LP and the UT for an object in GEO (Scenario 2) with initial EqOE covariance matrix $\mathbf{P} = \mathbf{A}\mathbf{A}^T$, where $\mathbf{A} = \text{diag}(\sigma_a = 2\,\text{km}, \sigma_h = 10^{-4}, \sigma_k = 10^{-4}, \sigma_p = 10^{-4}, \sigma_q = 10^{-4}, \sigma_\ell = 28'')$. The orbital state PDF is represented in Cartesian coordinates (red), EqOE (green), and J_2EqOE (blue), and is propagated for 30 days. For the PDF to be realistic (with 99.9% confidence), it is necessary that the value of the metric be within the shaded region

9. Although the underlying recommendation from Comment 1 is to use orbital element coordinates (i.e., EqOE or J_2EqOE) for representing and propagating orbital state uncertainty in space surveillance, in some applications, a Cartesian representation of the uncertainty might be required at a *future time*. To meet this requirement without loss of information in the process of converting an orbital state PDF in orbital elements to one in ECI, one can proceed in two ways. First, one can split the initial distribution (in Cartesian coordinates) at $t = t_0 = 0$ into a mixture of distributions and then propagate said mixture to a future epoch using an appropriate mixture filter (such as a Gaussian sum filter (Horwood and Poore 2011; Horwood et al. 2011; DeMars et al. 2012; Terejanu et al. 2008), if applicable). Alternatively, one can propagate the orbital state PDF in an orbital element coordinate system, and then transform said PDF to Cartesian coordinates at the final epoch time. Aristoff et al. (2014b) demonstrate how uncertainty in orbital elements can be faithfully (and efficiently) transformed to Cartesian space using Gaussian mixtures. This second approach is preferable since the computational cost of propagating a mixture of distributions is significantly greater than that of propagating a single distribution and then splitting it into a mixture (at the final epoch time).



(a) LP: CvM test statistic with J_2 dynamics

(b) LP: CvM test statistic with full dynamics

(c) UT: CvM test statistic with J_2 dynamics

(d) UT: CvM test statistic with full dynamics

Fig. 7 Statistical measure of uncertainty realism applied to the LP and the UT for an object in HEO (Scenario 3) with initial EqOE covariance matrix $\mathbf{P} = \mathbf{A}\mathbf{A}^T$, where $\mathbf{A} = \text{diag}(\sigma_a = 2\,\text{km}, \sigma_h = 10^{-4}, \sigma_k = 10^{-4}, \sigma_p = 10^{-4}, \sigma_q = 10^{-4}, \sigma_\ell = 28'')$. The orbital state PDF is represented in Cartesian coordinates (red), EqOE (green), and J_2EqOE (blue), and is propagated for 7 days. For the PDF to be realistic (with 99.9% confidence), it is necessary that the value of the metric be within the shaded region

We acknowledge that the initial covariances used in this study are contrived. Real-world covariances will generally be dense matrices; however, we cannot include results obtained on real data in this forum.

4 Discussion

Use of the J_2EqOE is not restricted to linearized propagation nor the unscented transform. As shown in Horwood et al. (2014a), the J_2EqOE can be combined with more sophisticated methods for uncertainty propagation, for example, the prediction step of the Gauss–von Mises filter (Horwood and Poore 2014), to further maintain uncertainty realism during uncertainty propagation. Indeed, for extreme uncertainties, the Gauss–von Mises filter is preferable due to its rigorous treatment of angular uncertainty and its computational efficiency (i.e., like the UT, only $2m + 1$ sigma points need to be propagated, where m is the length of the state vector).

As demonstrated in Fig. 4, the choice of coordinate system for representing the orbital state PDF has a profound impact when the covariance is considered "large." In this LEO example, the initial standard deviation along the semi-major axis (which coincides roughly with the radial direction) is 20 km. Such a large uncertainty can arise when creating a PDF from a single track of data (i.e., a short sequence of sensor observations) through an initial orbit determination process. Some applications (e.g., conjunction assessment) may require one to propagate this PDF and achieve covariance realism in order to obtain an accurate metric (e.g., probability of collision). Representing the PDF in J_2EqOE can significantly prolong covariance realism during propagation compared to Cartesian coordinates or EqOE.

Given the large impact of the choice of coordinate system on maintaining covariance realism during uncertainty propagation, it would be of interest to compare—using the statistical goodness-of-fit test described in Sect. 3.2—other analytic and semi-analytic orbital element theories such as those developed by Gim and Alfriend (2005) and Neelon et al. (1997), respectively. However, this is beyond the scope of the present study, and we leave it as future work.

5 Conclusions

The achievement of covariance and uncertainty realism is paramount to the success of several SSA functions including space surveillance tracking, conjunction assessment, sensor tasking, and maneuver detection. Without uncertainty realism, space objects can become lost, collisions can go undetected, and false alarms can lead to costly maneuvers. The J_2 equinoctial orbital elements, developed in this paper, are an example of a non-singular coordinate system in which the dynamics are less nonlinear and in which the orbital state PDF can be faithfully propagated for longer periods of time without a significant increase in computational cost. As demonstrated above using realistic force models and initial conditions, use of the J_2EqOE is especially beneficial when propagating objects in LEO and HEO, where the J_2 perturbation is most pronounced. Specifically, uncertainty realism can be maintained up to 28 times longer by using the J_2EqOE in place of Cartesian coordinates, and up to 14 times longer by using the J_2EqOE in place of the (Broucke and Cefola 1972) osculating EqOE.

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Appendix: Numerical example

To assist the reader in an implementation of the J_2EqOE to EqOE conversion (and its inverse), we provide details of how to recover the Keplerian orbital elements for the LEO example presented in Table 1 following all of the intermediate steps described in Sect. 2.

The following values are assumed for the gravitational parameter μ, Earth radius R, and J_2 coefficient:

$$\mu = 3.986004415\text{e+}05\,\text{km}^3/\text{s}^2$$
$$R = 6.378137\text{e+}03\,\text{km}$$

$$J_2 = 1.0826261738522\text{e-}03$$

The input J_2EqOE are the following:

$$n'' = 1.0487861550167\text{e}{-}03\,\text{s}^{-1} \qquad p'' = 6.6371635585001\text{e}{-}01$$
$$h'' = 1.4260897234681\text{e}{-}03 \qquad q'' = -3.2380704393120\text{e}{-}01$$
$$k'' = -9.2912382513865\text{e}{-}03 \qquad \ell'' = 4.8735182761240\text{e}{+}00$$

In what follows, all angular quantities have implied units of radians.

J_2EqOE to J_2IOE Conversion

$$I_1'' = 7.1294270251742\text{e}{+}03\,\text{km} \qquad I_4'' = 5.3390774990364\text{e}{-}01$$
$$I_2'' = 8.9420945754925\text{e}{-}03 \qquad I_5'' = -2.6047736914253\text{e}{-}01$$
$$I_3'' = -2.8982382142187\text{e}{-}03 \qquad I_6'' = 4.8735182761240\text{e}{+}00$$

J_2IOE to IOE Conversion: Step 1

$$a'' = 7.1294270251742\text{e}{+}03\,\text{km} \qquad h'' = 2.0246926216394\text{e}{+}00$$
$$e'' = 9.4000446883730\text{e}{-}03 \qquad g'' = 9.6460100038126\text{e}{-}01$$
$$I'' = 1.2721904014791\text{e}{+}00 \qquad l'' = 1.8842246541034\text{e}{+}00$$

J_2IOE to IOE Conversion: Step 2

$$k = 2.2020958264503\text{e}{+}04\,\text{km}^2 \qquad \gamma = 4.3323841440301\text{e}{-}04$$
$$\eta = 9.9995581860393\text{e}{-}01 \qquad \gamma' = 4.3331498717247\text{e}{-}04$$
$$\theta = 2.9418810951484\text{e}{-}01$$

J_2IOE to IOE Conversion: Step 3

$$E'' = 1.8931405532060\text{e}{+}00$$

J_2IOE to IOE Conversion: Step 4

$$r'' = 7.1506573807183\text{e}{+}03\,\text{km} \qquad f'' = 1.9020433345198\text{e}{+}00$$

J_2IOE to IOE Conversion: Step 5

$$a = 7.1366000000000\text{e}{+}03\,\text{km} \qquad \delta h = -1.1070091328349\text{e}{-}04$$

J_2IOE to IOE Conversion: Step 6

$$l + g + h = 4.8729592715682\text{e}{+}00$$

J_2IOE to IOE Conversion: Step 7

$$\nu_1 = -9.7268781528180\text{e}{-01} \qquad \delta e = 8.1222972039185\text{e}{-05}$$
$$\nu_2 = -9.5884220957866\text{e}{-01} \qquad e''\delta l = -4.0701787616281\text{e}{-04}$$
$$\nu_3 = -9.6354316647622\text{e}{-01}$$
$$\nu_4 = 1.9910139555114\text{e}{+00}$$

J_2IOE to IOE Conversion: Step 8

$$\delta I = 1.5460976151790\text{e}{-04} \qquad \sin\left(\tfrac{1}{2}I''\right)\delta h = -6.5762859846059\text{e}{-05}$$

J_2IOE to IOE Conversion: Step 9

$$I_1 = 7.1366000000000\text{e}{+03}\,\text{km} \qquad I_4 = 5.3399247411729\text{e}{-01}$$
$$I_2 = 9.1448530009497\text{e}{-03} \qquad I_5 = -2.6044553167591\text{e}{-01}$$
$$I_3 = -2.5360921889622\text{e}{-03} \qquad I_6 = 4.8729592715682\text{e}{+00}$$

IOE to EqOE Conversion

$$a = 7.1366000000000\text{e}{+03}\,\text{km} \qquad p = 6.6385958338730\text{e}{-01}$$
$$h = 1.0413786122339\text{e}{-03} \qquad q = -3.2378595304974\text{e}{-01}$$
$$k = -9.4326894672662\text{e}{-03} \qquad \ell = 4.8729592715682\text{e}{+00}$$

IOE to Keplerian Orbital Elements Conversion (Optional)

$$a = 7.1366000000000\text{e}{+03}\,\text{km} \qquad \Omega = 2.0245819323134\text{e}{+00}$$
$$e = 9.4900000000000\text{e}{-03} \qquad \omega = 1.0070549784028\text{e}{+00}$$
$$i = 1.2723450247039\text{e}{+00} \qquad M = 1.8413223608519\text{e}{+00}$$

These Keplerian orbital elements are precisely those for the LEO object in Table 1. Note that the angular quantities (i, Ω, ω, M) in the above are given in radians whereas the table converts them to degrees.

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